(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION

(Course Name: B. Tech) (Semester: V)

(October, 2025) OFF LINE mode

Subject Code: BAI 305	Subject: Deep Learning -I
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)
	a) Explain the structure of a McCulloch–Pitts neuron with the help of a diagram.	
	b) "Perceptron model considers the non-linear combination of inputs." Verify the statement.	
	c) Differentiate between overfitting and underfitting in neural networks?	
	d) Explain gradient descent and its purpose in training neural networks.	
	UNIT I	
Q2	**Attempt any two parts**	(5*2=10)
	a) Describe the Perceptron Learning Algorithm (PLA) in context of perceptron model along its convergence proof.	
	b) Why does the perceptron fail for non-linearly separable problems? Explain with suitable example.	
	c) Explain the thresholding function? What are the advantages of thresholding function over logistic function?	
	UNIT II	
Q3	**Attempt any two parts**	(5*2=10)
	a) Explain the architecture and working of a Multi-Layer Perceptron (MLP). Explain the architectural design parameters for a MLP.	
	b) Discuss the different methods to prevent overfitting of deep learning models.	
	c) Describe the Backpropagation algorithm of MLP with an example.	

Scanned with OKEN Scanner